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Crime, correction, education and welfare in the U.S. – What role does the government play?

Devika Hazra*, Jose Aranzazu

California State University, Los Angeles, USA

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Abstract

As federal, state and local governments continue to allocate a significant share of their resources to law enforcement and correctional spending, concerns have risen that spending in education and welfare is declining. With fiscal pressure in the United States mounting, it is important to determine the effectiveness of public spending in deterring crime. This paper compares the effectiveness of the impact of government spending on welfare and education with that of law enforcement and correction on crime. Using panel data from 50 U.S. states over a time period of 1994–2014, results of linear regression with panel corrected standard errors as well as GMM estimation reveals that public welfare and education spending can potentially lower violent and property crime rates but law enforcement spending can only deter property crime. However, correctional spending can exacerbate both types of crimes. There is little to no evidence of the presence of crowding out of one category of spending by another. This results in the policy implication that more resources be allocated towards welfare and education programs.
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* Corresponding author.
E-mail address: dhazra@calstatela.edu (D. Hazra).

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1. Introduction

High crime rates have been an impediment to society's progress since time immemorial. While crime rates are lower compared to the 1960s and continues to exhibit a declining trend (as shown in Fig. 1), expenditure per capita by state governments on law enforcement and correction continues to soar.

A relatively large group of economists, social scientists and policymakers contend that a reduction in crime can be facilitated by additional expenditures on law enforcement resources. The idea of deterrence effect, that is, increasing police presence leads to an increase in the probability of being caught which tends to lower the net expected utility of crime can be attributed to Becker (1968). However, given the limited nature of resources, such increase in crime-related spending could entail rising opportunity costs – this cost being the value of forgone alternatives to which the resources could have been allocated. In an environment of escalating fiscal pressure, the efficacy of public expenditures allocated towards deterring crime emerges as a significant issue. The aim of this paper is to analyze the effectiveness of public expenditures in curbing crime rates across the United States. Between 1982 and 2007, expenditure on nation's police protection, adjusted for inflation increased by a staggering 171% and that on corrections, which was mainly concentrated at the state level, varied between 59% and 63%.[1] Such rising crime-related spending has not only been a bone of contention among proponents of public education and public welfare as well as policy makers but also the epicenter of more recent "Defund the Police[2]" initiative. The common conjecture is that if characteristics, such as education, health and welfare, that are connected to criminal behavior can be influenced, it is possible to not only counter crime but also increase human and social capital in

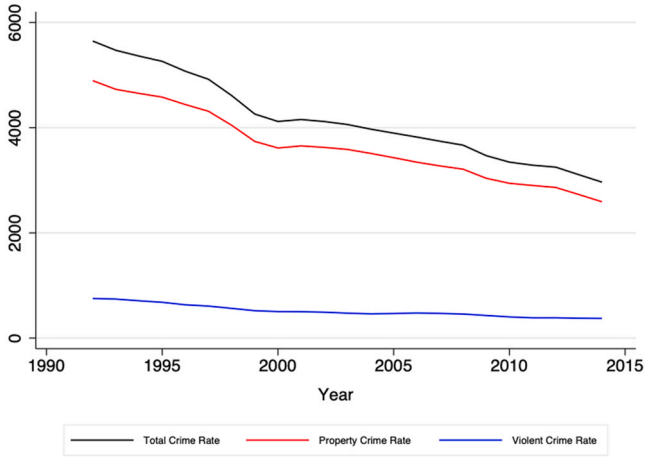

Fig. 1. Trend in crime rate.

[1] https://www.bjs.gov/content/pub/pdf/jee8207st.pdf.

[2] "Defund the police" means reallocating or redirecting funding away from the police department to other government agencies funded by the local municipality. (Source: Brookings)

the economy. This paper investigates whether government expenditures on education and welfare have a greater impact on crime rates relative to crime-related government expenditures.

A concern often expressed is that a focus on public expenditures to lower crime tends to be myopic – yielding results only in the short run. However, policies that are oriented towards building human capital via public expenditures on education and health tend to exhibit long run results by lowering crime not only in the present but also in the future. This in turn, leads to a debate regarding the relationship between expenditures on health, education and welfare vis-à-vis expenditures on law enforcement and correction. It is often reckoned that such categories of public expenditure compete with each other such that one program gains at the expense of others. However, there is an alternative hypothesis which conjectures that these expenditures are unrelated in that there could be an increase in both types of spending where such spending might be accompanied by either a rise in taxes or an issuance of public debt. This paper examines if indeed one category of public expenditure such as law enforcement and correction crowds out other categories of public spending, namely, education, and welfare using panel data spanning 20 years across 50 US states.

There is extensive literature relating to the impact of education on crime. However, the same cannot be said for the relationship between education expenditures on crime as well as expenditures on public welfare on crime. In particular, the literature that explores the aforementioned crowding out effect is almost non-existent (Marlow & Shiers, 1999, 2001). Increased spending in public schools and colleges lowers the burden of the cost of education borne by individuals. This renders education as more affordable and hence, accessible. Several studies have been undertaken to examine the causal effects of education on crime. One idea behind such effect is that education raises the opportunity cost of committing crimes. Higher education increases the probability of acquiring work causing a greater decrease in wages if the person is incarcerated (Lochner & Moretti, 2004). Machin, Marie, and Vujić (2011) corroborate that there is a significant potential to reduce crime by investing in human capital and schooling. Additional studies that further establish this causal relationship include Åslund, Grönqvist, Hall, and Vlachos (2018) which contend that increasing access to secondary education lowers property crime and Bell, Costa, and Machin (2016) which concluded that longer periods of compulsory education results in lower crime. Further, Usher (1997) contends that public schooling, rather than private schooling (even with voucher scheme) exhibits a civic externality of crime deterrence. Chalfin and Deza (2019) show that a longer period of compulsory education for parents will decrease crimes committed by their children, thereby, establishing an intergenerational effect of devoting greater resources to education on crime. On examining the relationship between shadow economy and public expenditure on education, Berrittella (2015) finds that an increase in the latter leads to lowering of the former. Other examples from education literature include Sabates and Feinstein (2008) and Sabates (2009) and from criminology literature include Farrington (1999). The consensus in the literature that educational attainment can be a crime deterrent establishes the ground for probing the question of whether government spending on education rather than law-enforcement lower crime rates – a question this paper seeks to address.

Although there is a dearth of analysis that examine the relationship between crime rates and social welfare spending, there is consensus in the existing literature regarding the existence of a negative relationship between the two despite early disagreements. Studies by Burek (2006), D'Alessio, Stolzenberg, and Eitle (2002) and Worrall (2005) argue in favor of a positive relationship between welfare spending and crime rates; and that by Chamlin, Cochran, and Lowenkamp (2002) find the lack of any association. However, these analyzes were criticized on

the grounds of their cross-sectional nature and failure to capture temporal variations. Worrall (2009)'s re-evaluation of his original study supported the negative relationship between social welfare spending and crime. This viewpoint holds that a lack of legitimate opportunity emanating from discrimination and inequities in education, employment and housing, accompanied by stress and subsequent frustration may induce specific groups of people to engage in criminal activity. Consequently, welfare assistance is tantamount to redress the imbalance of opportunities within society and palliate the tensions that cause crime. Studies by Cebula (1991), Davey (1995), DeFronzo (1983, 1997), Defronzo and Hannon (1998), Devine, Sheley, and Smith (1988), Foley (2011), Pratt & Godsey, (2002), Meloni (2014), Monte and Lewis (2011), Sampson, Wilson, Hagan, and Peterson (1995), Savage, Bennett, and Danner (2008), Thompson and Uggen (2013) and Zhang (1997) reveal that welfare assistance to the poor is negatively associated with violent crimes, including homicide as well as property crimes, including theft. An analysis by Fishback, Johnson, and Kantor (2010) suggest a negative relationship between relief spending and property crime. Limiting the availability of free time in the form of work relief tends to reduce crime. Additionally, Rudolph and Starke (2020) found that social support via generous unemployment benefits is beneficial in curbing crime. Despite studies demonstrating the crime-reducing effect of social welfare programs, public discourse and policy in the United States is reflective of the rhetoric that such welfare programs foster, rather than reduce, criminal activity. Brown (2016), Cullen, Wright, and Chamlin (1999), Garland (2001) and Wacquant (2009) corroborate a punitive response to crime reduction. The crux of this hypothesis is that aid to the poor propagates dependency by weakening personal initiative wherein it advocates for stricter criminal prosecution. This paper explores the relationship between crime rates and welfare spending by the government to shed light on the crime-welfare relationship vis-à-vis crime-law enforcement spending in the United States using more recent panel data.

Empirical evidence on the deterrence effect of crime theorized by Becker (1968) is widely varied. Cameron (1988), Eck and Maguire (2000), Greenwood and Wadycki (1973) and Marvell and Moody (1996), among others either found positive correlation or no evidence of association between law enforcement and crime rates. Additionally, Pogue (1975) established that difference in crime rates between cities cannot be attributed to differences in resources allocated towards law enforcement. Analysis by Kollias, Mylonidis, and Paleologou (2013) also revealed the lack of association between public order outlays and crime rate. Such conclusion may be attributed to inefficient use of resources or be indicative of the presence of an array of other factors that also affect both crime rates and policing. In fact, Fisher (1996), Levitt (2002a, 2002b), Marvell and Moody (1996) and Nagin (1998) point towards model misspecification and/or the presence of endogeneity problem in these studies. In their studies, police presence and crime exhibit negative correlation. Wilson and Boland (1978) find that both police resources and police presence affect crime rates negatively. More recent studies by Chalfin and McCrary (2013) and Lin (2009) further corroborate this result. In this paper we find contradictory evidence – while an increase in spending on law-enforcement leads to lowering of property crime rate, increase in correctional spending positively affects both property and violent crime rates.

The remainder of the paper is organized as follows. Section 2 provides a description of the data and presents basic crime statistics, Section 3 discusses methodology, Section 4 discusses the results and conducts various robustness checks and Section 5 concludes.

2. Data description

This paper uses data on violent crime and property crime for all 50 U.S. states for a span of twenty years: 1992–2014. Property crimes include burglary, larceny-theft, and motor vehicle theft. Violent crimes include murder/non-negligent manslaughter, rape, robbery, and aggravated assault. Crime data was collected from the FBI's Uniform Crime Reporting (UCR) Program. Fig. 1 shows that there is a downward trend in crime rate.

State level expenditures on welfare, education, law enforcement and correction were collected from the state financial reports that are available at U.S. census bureau. Expenditures on public welfare include means-tested programs, such as Medicaid, Temporary Assistance for Needy Families, and Supplemental Security Income. Expenditures on education include expenditures incurred by states on schools, colleges, educational programs for adults, and educational programs for veterans. Expenditures on correction consist of state-level spending on confinement and correction for adults and minors that commit crimes. Expenditures on law enforcement is the amount spent by states on the preservation of the law, which includes police patrols, detention, and traffic safety.

Data for additional control variables such as poverty rate, educational attainment, age dependency ratio, population density, population in metro area, school enrollment rate as well as total population was also collected from the census bureau. For Gini Index of income inequality (GINI) Internal Revenue Services (IRS) data-based calculations by Frank (2014)[3] were used. Data for nominal and real GDP were extracted from the Bureau of Economic Analysis. Unemployment rate data was obtained from the Bureau of Labor Statistics. Data for consumer price index (CPI) was collected from the Federal Reserve Bank of St. Louis and was used to calculate the annual inflation rate.[4] Additionally, data on number of law enforcement and correctional employees were available at the Bureau of Justice Statistics. Data on Presidential, Senate and House election results were obtained from MIT Election Data and Science Lab (Massachusetts Institute of Technology) which was then used to determine whether a state is majority Democrat or Republican.

3. Empirical strategy

There are four main independent variables of interest: education expenditure as a percentage of GDP (Eduperc), public welfare expenditure as a percentage of GDP (Welfperc), law enforcement expenditure as a percentage of GDP (Lawperc), and correctional expenditure as a percentage of GDP (Corrperc). Since there is a possibility of a bi-directional relationship between these expenditure categories and crime, that is, the amount of expenditures allocated towards crime may depend on the number of crime incidences, we use lagged values of these variables. Additionally, log of real GDP per capita (RGDpc), unemployment rate (UnRate), inflation rate (InfRate), poverty rate (PovRate), Gini index of income inequality (GINI), age dependency ratio (ADR), log of population density (PopDen), percentage of population living in metropolitan area, majority political orientation of the state (Politic) and percentage of population 25 years or older with at least a bachelor's degree (EduAtt) are used as control variables. Violent crime rate (VCR) and property crime rate (PCR) are the two independent

[3] https://www.shsu.edu/eco_mwf/inequality.html.
[4] Data for CPI is not available at the state level.

variables, where both measure the number of crimes per 100,000. Table 1 shows the summary statistics of the underlying variables.

First, we use augmented Dickey-Fuller unit root test to examine all dependent and independent variables. As shown in Table 2, the results indicate that none of the variables have unit root at level.

A Lagrange-Multiplier test for autocorrelation, as shown in Table 3 reveals that there is first order autocorrelation in the panel data:

Next, we use a Likelihood Ratio test to check for heteroskedasticity. The result of the test displayed in Table 4 indicates the presence of panel-level heteroskedasticity:

Further, Pesaran's test of cross-sectional independence is conducted. As evident from Table 5, the null hypothesis that the residuals are not correlated cannot be rejected:

Since there is evidence of autocorrelation, heteroskedasticity and cross-sectional dependence, an OLS estimation approach would be biased and inconsistent. Therefore, first, a linear regression with Beck and Katz (1995) panel-corrected standard errors is implemented. With CR_{it}^{x} being Crime Rate for State i at time t and x indicates either Property Crime Rate or Violent Crime Rate, respectively, this model can be written as shown in Eq. (1):

$$\begin{aligned} CR_{it}^{x} = {} & \beta_0^a + \beta_1^a Eduperc_{it-1} + \beta_2^a Welfperc_{it-1} + \beta_3^a Lawperc_{it-1} \\ & + \beta_4^a Corrperc_{it-1} + \beta_5^a lnRGDpc_{it} + \beta_6^a UnRate_{it} + \beta_7^a PovRate_{it} \\ & + \beta_8^a InfRate_{it} + \beta_9^a GINI_{it} + \beta_{10}^a ADR_{it} + \beta_{11}^a lnPopDen_{it} + \beta_{12}^a metro_{it} \\ & + \beta_{13}^a Politic_{it-1} + \beta_{14}^a EduAt_{it} + \varepsilon_{it}^a \end{aligned} \tag{1}$$

Second, we conduct a two-step Generalized Method of Moments (GMM) estimation using a system of equations since linear regression models might not provide a comprehensive picture of the crime and government spending nexus. As mentioned previously, one hypothesis

Table 1
Descriptive statistic.

Variables	Mean	Standard deviation	Min	Max
Total Crime Rate	3970.64	1175.05	1623.62	8358.24
Property Crime Rate	3536.01	1029.03	1524.35	7500.12
Violent Crime Rate	434.63	211.96	66.91	1207.20
Education Expenditure per capita	3.99	1.07	1.63	8.77
Welfare Expenditure per capita	2.79	0.89	1.00	6.02
Law Enforcement Expenditure per capita	0.10	0.05	0.01	0.38
Correctional Expenditure per capita	0.33	0.09	0.10	0.68
Real GDP per capita	43,317.6	12,080.0	18,145.9	80,035.8
Unemployment Rate	5.68	1.90	2.30	13.7
Inflation Rate	2.41	0.90	-0.36	3.84
Poverty Rate	12.7	3.58	4.50	26.4
Gini Index of Income Inequality	0.59	0.036	0.52	0.71
Age Dependency Ratio	60.3	5.36	41.8	81.5
Population Density	155.7	195.7	0.89	1024.8
% Population in metro area	82.4	28.8	11.1	851.5
Political Orientation	0.48	0.50	0	1
% above 25 years with bachelor's degree	24.8	5.78	7.50	41.2
School Enrollment Rate	91.7	3.39	77.9	103.2
Police per capita	41.0	18.4	13.4	120.8
Correctional Employees per capita	159.1	85.5	16.8	921.8
Number of observations: 1150				

Table 2
Results of unit root test.

Variables	Augmented Dickey-Fuller test statistics
Property Crime Rate (CR_{it}^{PCR})	121.20*
Violent Crime Rate (CR_{it}^{VCR})	278.92***
Education Expenditure per capita ($Eduperc_{it}$)	344.11***
Welfare Expenditure per capita ($Welfperc_{it}$)	206.38***
Law Enforcement Expenditure per capita ($Lawperc_{it}$)	319.43***
Correctional Expenditure per capita ($Corperc_{it}$)	348.64***
Real GDP per capita ($RGDpc_{it}$)	425.03***
Unemployment Rate ($UnRate_{it}$)	469.50***
Inflation Rate ($InfRate_{it}$)	713.28***
Poverty Rate ($PovRate_{it}$)	379.64***
Gini Index of Income Inequality ($GINI_{it}$)	248.47***
Age Dependency Ratio (ADR_{it})	404.18***
Population Density ($PopDen_{it}$)	221.24***
% Population in metro area ($Metro_{it}$)	277.47***
Political Orientation ($Polit_{it}$)	265.62***
% above 25 years with bachelor's degree ($EduAtt_{it}$)	420.37***
School Enrollment Rate ($ScEnRate_{it}$)	266.99***
Police per capita ($Polpc_{it}$)	285.70***
Correctional Employees per capita ($CorrOffpc_{it}$)	452.79***

Notes: Statistical significance level: *** 1%; ** 5%; *10%.

Table 3
Wooldridge test for autocorrelation in panel data.

H0: no first order autocorrelation	
Property crime rate	Violent crime rate
$F(1, 49) = 244.363$	$F(1, 49) = 105.830$
$Prob > F = 0.00$	$Prob > F = 0.00$

Table 4
Likelihood ratio test.

Property crime rate	Violent crime rate
LR $\chi^2(49) = 519.54$	LR $\chi^2(49) = 590.87$
Prob > $\chi^2 = 0.000$	Prob > $\chi^2 = 0.000$

Table 5
Pesaran's test of cross-sectional independence.

	Property crime rate	Violent crime rate
Pesaran's test of cross-sectional independence	31.353	16.579
Pr	0.000	0.000
Average Absolute value of the off-diagonal elements	0.383	0.407

governing the literature is that crime-related spending, education spending, and welfare spending compete with each other. Alternatively, it has also been hypothesized that various categories of spending are unrelated and higher spending on any category may be accompanied by issuance of public debt or taxation. Generalized Method of Moments (GMM) allows not only for correlation between right-hand side variables and errors, but also for correlation across residuals, autocorrelation in the residuals, and heteroskedasticity. Therefore, we estimate a five-equation model of crime rate, welfare spending, education spending, law-enforcement spending and correctional spending. The exogenous variables of each equation were used as instruments for the respective equations leading to an exactly identified GMM system of equations. Eqs. (1)–(5) was estimated using GMM with cluster-robust variance-covariance matrix. The additional exogenous variables used are number of police per capita (PolPC) and number of correctional officers per capita (CorrOffpc).

$$\begin{aligned} Welfperc_{it-1} = {} & \beta_0^c + \beta_1^c Eduperc_{it-1} + \beta_2^c Lawperc_{it-1} \\ & + \beta_3^c Corrperc_{it-1} + \beta_4^c lnRGDpc_{it-2} \\ & + \beta_5^c PovRate_{it-2} + \beta_6^c UnRate_{it-2} \\ & + \beta_7^c Politic_{it-1} + \varepsilon_{it}^c \end{aligned} \tag{2}$$

$$\begin{aligned} Eduperc_{it-1} = {} & \beta_0^b + \beta_1^b Welfperc_{it-1} + \beta_2^b Lawperc_{it-1} \\ & + \beta_3^b Corrperc_{it-1} + \beta_4^b lnRGDpc_{it-2} \\ & + \beta_5^b SchEnRate_{it-1} + \beta_6^b Politic_{it-1} + \varepsilon_{it}^b \end{aligned} \tag{3}$$

$$\begin{aligned} Lawperc_{it-1} = {} & \beta_0^d + \beta_1^d Eduperc_{it-1} + \beta_2^d Welfperc_{it-1} \\ & + \beta_3^d Corrperc_{it-1} + \beta_4^d lnRGDpc_{it-2} \\ & + \beta_5^d Polpc_{it-2} + \beta_6^d CR_{it-2}^{Total} \\ & + \beta_7^d Politic_{it-1} + \varepsilon_{it}^d \end{aligned} \tag{4}$$

$$\begin{aligned} Corrperc_{it-1} = {} & \beta_0^e + \beta_1^e Eduperc_{it-1} \\ & + \beta_2^e Welfperc_{it-1} \\ & + \beta_3^e Lawperc_{it-1} + \beta_4^e lnRGDpc_{it-2} \\ & + \beta_5^e CorrOffpc_{it-2} \qquad\qquad + \beta_6^e CR_{it-2}^{Total} \\ & + \beta_7^e Politic_{it-1} + \varepsilon_{it}^e \end{aligned} \tag{5}$$

4. Results and discussion

The results of linear regression with panel-corrected standard errors is shown in column 1 of Table 6 and Table 7 for property crime rate and violent crime rate, respectively. Percent of education expenditures as well as welfare expenditures has a crime-reducing effect for both categories. A one percent point increase in Eduperc, will lead to a reduction by 0.022% and 0.082% points of property crime rate and violent crime rate, respectively. An increase in welfare

expenditures by one percent point, leads to a reduction in property crime rate and violent crime rate by 0.074% and 0.117% points, respectively. Notably, both education spending as well as welfare spending has a relatively greater impact on violent crime rate compared to property crime rate. Second, the marginal impact of welfare spending relative to that of education spending is greater for violent crime as well as property crime. Welfare spending, whether by providing direct means of livelihood by way of income transfers, food stamps, etc. or by generating work for the unemployed (work relief), reduces the probability of committing crimes. However, in spite of educational attainment exhibiting the ability to curb crime, the effect of education spending by the government manifests via an indirect channel and therefore, the impact may not be as strong as that of welfare spending. When there is increase in education expenditures, it lowers the cost of attending school/college for lower-income households and increases the probability of attaining higher education which in turn provides opportunities to acquire higher paying jobs and lowers the return on crime.

The impact of expenditures on law enforcement is significant only for property crime rate. A unit increase in percent spending in law enforcement expenditures leads to a 0.48% point reduction in property crime rate. Since increasing expenditures on law enforcement contributes to increased police patrols and surveillance, it makes committing burglaries and thefts more difficult. The increase in the probability of getting caught, decreases the likelihood of committing a crime. Notably, the crime-reducing impact of law enforcement expenditures is higher relative to that of welfare and education expenditures on property crime rate. Surprisingly, an increase in correctional expenditures leads to increases in both types of crime rates. For every unit increase in Corrperc_{it}, violent and property crime rate increases by 0.743% and 0.423% points, respectively. This variable, of all four categories of spending, has the greatest impact on violent crime rate, but exerts an undesirable impact. There can be several reasons for this positive effect on crime rates. Correctional spending involves confinement and correction of both minors and adults. Increasing spending on prison and confinement facilities and officers may allow for a higher rate of incarceration, it may not have the desired impact on those imprisoned. Rather, it can diminish the development of human capital and hence, future labor market outcomes of those incarcerated, and in turn, pave the path for increased criminal activities. Further, it can exert an adverse effect on communities by negatively affecting social relationships and damaging networks of the families of those imprisoned in an undesirable manner who may be forced, under circumstances, to undertake criminal activities.

Among the macroeconomic variables, while real GDP per capita lowers property crime rate, it has no effect on violent crime rate. An increase in economic prosperity brings forth higher income earning opportunities which tends to lower the incentive to commit property crimes. Neither unemployment rate nor inflation rate is significant in explaining variations in either types of crime rates. While the socioeconomic variable poverty rate leads to increase in crime rates, educational attainment exhibits a crime-reducing effect on both types of crime rates. Poverty leads to deprivation, and possibly creates a sense of desperation to obtain means of livelihood. When the expected return on legal market activities is low, low-income individuals may resort to theft, robbery, etc. Additionally, it may also lead to increase in violent crimes, such as armed robbery. Merton (1938) study depicts individuals in the lower income group become alienated from society and tend to commit crimes if they are unsuccessful in attaining material attributes of success. Educational attainment exhibits a crime-deterring effect. As discussed earlier, existing literature on the effect of educational attainment and crime conjectures that higher the educational attainment, higher is the expected earnings from legal market activities and thus, higher is the opportunity cost of committing crime.

Table 6
Results of regression analysis I.

Dependent variable: property crime rate							
	Linear regression, with PCSE		Generalized method of moments for a system of equations				
Variables	Main model (CR_{it}^{PCR})	With year fixed effects (CR_{it}^{PCR})	Main model (CR_{it}^{PCR})	% Welfare expenditure ($Welfperc_{it-1}$)	% Education expenditure ($Eduperc_{it-1}$)	% Law enforcement expenditure ($Lawperc_{it-1}$)	% Correctional expenditure ($Corrperc_{it-1}$)
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
% Education Expenditure ($Eduperc_{it-1}$)	-0.022** (0.010)	-0.018*(0.009)	0.011 (0.029)	0.319*** (0.072)		0.008 (0.005)	0.015** (0.008)
% Welfare Expenditure ($Welfperc_{it-1}$)	-0.074*** (0.010)	-0.046*** (0.009)	-0.155*** (0.027)		0.480*** (0.104)	0.007 (0.004)	0.006 (0.008)
Law Enforcement Expenditure ($Lawperc_{it-1}$)	-0.483*** (0.150)	-0.522*** (0.134)	0.200 (0.487)	4.32** (1.717)	5.559** (2.474)		0.487** (0.191)
Correctional Expenditure per capita ($Corperc_{it-1}$)	0.423*** (0.094)	0.369*** (0.083)	0.798*** (0.255)	-1.60** (0.707)	1.786 (1.124)	0.104** (0.042)	
Real GDP per capita ($RGDpc_{it}$)	-0.166*** (0.060)	0.015 (0.045)	-0.316*** (0.075)				
Real GDP per capita ($RGDPC_{it-2}$)				0.269 (0.138)	-0.658*** (0.170)	0.010 (0.010)	0.060*** (0.022)
Unemployment Rate ($UnRate_{it}$)	-0.002 (0.007)	0.008** (0.004)	0.005 (0.011)				
Unemployment Rate ($UnRate_{it-2}$)				0.105*** (0.024)			
Poverty Rate ($PovRate_{it}$)	0.004*** (0.001)	0.004*** (0.001)	0.017** (0.007)				
Poverty Rate ($PovRate_{it-2}$)				0.043*(0.026)			
Inflation Rate ($InfRate_{it}$)	0.004 (0.007)	0.092*** (0.011)	0.035*** (0.007)				
Income Inequality ($GINI_{it}$)	-0.517 (0.316)	0.150 (0.188)	-0.848 (0.546)				
Age Dependency Ratio (ADR_{it})	0.0001 (0.002)	0.005*** (0.001)	0.0003 (0.004)				

(*continued on next page*)

Table 6 (*continued*)

Dependent variable: property crime rate							
	Linear regression, with PCSE		Generalized method of moments for a system of equations				
Variables	Main model (CR_{it}^{PCR})	With year fixed effects (CR_{it}^{PCR})	Main model (CR_{it}^{PCR})	% Welfare expenditure ($Welfperc_{it-1}$)	% Education expenditure ($Eduperc_{it-1}$)	% Law enforcement expenditure ($Lawperc_{it-1}$)	% Correctional expenditure ($Corrperc_{it-1}$)
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Population Density ($PopDen_{it}$)	-0.011 (0.013)	-0.011 (0.008)	0.014 (0.022)				
% Population in metro area ($Metro_{it}$)	-0.00003 (0.0001)	0.0001 (0.002)	0.001 (0.0005)				
% with bachelor's degree ($EduAtt_{it}$)	-0.009*** (0.003)	-0.004** (0.002)	-0.009** (0.005)				
Political ($Polit_{it}$)	-0.029*** (0.008)	-0.020*** (0.007)	-0.029*(0.038)	0.306** (0.138)	-0.280*(0.161)	0.005 (0.005)	0.005 (0.014)
School Enrollment Rate ($ScEnRate_{it-2}$)					0.081*** (0.020)		
Total Crime Rate ($TCrime_{it-2}$)						0.002 (0.014)	0.109*** (0.033)
Police per capita ($Polpc_{it-2}$)						0.065*** (0.009)	
Correctional Officers per capita ($CorrOffpc_{it-2}$)							0.055*** (0.014)
Constant	10.648*** (0.642)	8.026*** (0.544)	11.920*** (0.843)	-2.538 (2.111)	1.175 (2.829)	-0.340 (0.207)	-1.610*** (0.434)
Observations	1100	1100	1029	1029	1029	1029	1029
R-squared	0.998	0.998					

Standard errors in parentheses.
Robust standard errors in parentheses.
*** $p < 0.01$.
** $p < 0.05$.
* $p < 0.1$.

Table 7
Results of regression analysis II.

Dependent variable: violent crime rate							
	Linear regression, with PCSE		Generalized method of moments for a system of equations				
Variables	Main Model (CR_{it}^{VCR}) (1)	With year fixed effects (CR_{it}^{VCR}) (2)	Main model (CR_{it}^{VCR}) (3)	% Welfare expenditure ($Welfperc_{it-1}$) (4)	% Education expenditure ($Eduperc_{it-1}$) (5)	% Law enforcement expenditure ($Lawperc_{it-1}$) (6)	% Correctional expenditure ($Corrperc_{it-1}$) (7)
% Education Expenditure ($Eduperc_{it-1}$)	-0.082*** (0.014)	-0.05*** (0.014)	-0.107* (0.057)	0.319*** (0.072)		0.008 (0.005)	0.015** (0.008)
% Welfare Expenditure ($Welfperc_{it-1}$)	-0.117*** (0.017)	-0.099*** (0.016)	-0.218*** (0.063)		0.480*** (0.104)	0.007 (0.004)	0.006 (0.008)
Law Enforcement Expenditure ($Lawperc_{it-1}$)	0.312 (0.216)	-0.091 (0.220)	1.164 (1.011)	4.32** (1.717)	5.559** (2.474)		0.487** (0.191)
Correctional Expenditure per capita ($Corperc_{it-1}$)	0.743*** (0.135)	0.873** (0.128)	1.779*** (0.504)	-1.60** (0.707)	1.786 (1.124)	0.104** (0.042)	
Real GDP per capita ($RGDPC_{it}$)	-0.038 (0.062)	0.489*** (0.093)	0.173 (0.187)				
Real GDP per capita ($RGDPC_{it-2}$)				0.269 (0.138)	-0.658*** (0.170)	0.010 (0.010)	0.060*** (0.022)
Unemployment Rate ($UnRate_{it}$)	-0.002 (0.007)	0.014 (0.009)	0.030 (0.022)				
Unemployment Rate ($UnRate_{it-2}$)				0.105*** (0.024)			

(*continued on next page*)

Table 7 (*continued*)

Dependent variable: violent crime rate							
	Linear regression, with PCSE		Generalized method of moments for a system of equations				
Variables	Main Model (CR_{it}^{VCR}) (1)	With year fixed effects (CR_{it}^{VCR}) (2)	Main model (CR_{it}^{VCR}) (3)	% Welfare expenditure ($Welfperc_{it-1}$) (4)	% Education expenditure ($Eduperc_{it-1}$) (5)	% Law enforcement expenditure ($Lawperc_{it-1}$) (6)	% Correctional expenditure ($Corrperc_{it-1}$) (7)
Poverty Rate ($PovRate_{it}$)	0.007*** (0.002)	0.010*** (0.002)	0.048*** (0.015)				
Poverty Rate ($PovRate_{it-2}$)				0.043* (0.026)			
Inflation Rate ($InfRate_{it}$)	0.001 (0.006)	0.136*** (0.020)	0.050*** (0.015)				
Income Inequality ($GINI_{it}$)	0.337 (0.373)	0.910** (0.393)	0.636 (0.935)				
Age Dependency Ratio (ADR_{it})	-0.001 (0.002)	0.006*** (0.002)	0.0001 (0.991)				
Population Density ($PopDen_{it}$)	0.139*** (0.0189	0.122*** (0.018)	0.094* (0.053)				
% Population in metro area ($Metro_{it}$)	0.0001 (0.0001)	0.0001 (0.019)	0.001 (0.001)				
% with bachelor's degree ($EduAtt_{it}$)	-0.016*** (0.003)	-0.017*** (0.003)	-0.031*** (0.010)				
Political ($Polit_{it}$)	-0.038*** (0.014)	-0.037*** (0.012)	-0.015 (0.076)	0.306** (0.138)	-0.280*(0.161)	0.005 (0.005)	0.005 (0.014)

(*continued on next page*)

Table 7 (*continued*)

Dependent variable: violent crime rate							
	Linear regression, with PCSE		Generalized method of moments for a system of equations				
Variables	Main Model (CR_{it}^{VCR}) (1)	With year fixed effects (CR_{it}^{VCR}) (2)	Main model (CR_{it}^{VCR}) (3)	% Welfare expenditure ($Welfperc_{it-1}$) (4)	% Education expenditure ($Eduperc_{it-1}$) (5)	% Law enforcement expenditure ($Lawperc_{it-1}$) (6)	% Correctional expenditure ($Corrperc_{it-1}$) (7)
School Enrollment Rate ($ScEnRate_{it-2}$)					0.081*** (0.020)		
Total Crime Rate ($TCrime_{it-2}$)						0.002 (0.014)	0.109*** (0.033)
Police per capita ($Polpc_{it-2}$)						0.065*** (0.009)	
Correctional Officers per capita ($CorrOffpc_{it-2}$)							0.055*** (0.014)
Constant	6.375*** (0.714)	-0.307 (1.084)	-3.415 (2.115)	-2.538 (2.111)	1.175 (2.829)	-0.340 (0.207)	-1.610*** (0.434)
Observations	1100	1100	1029	1029	1029	1029	1029
R-squared	0.987	0.991					

Standard errors in parentheses.
Robust standard errors in parentheses.
*** $p < 0.01$.
** $p < 0.05$.
* $p < 0.1$.

Demographic variable log of population density has a positive impact on violent crime rate. A 1% increase in population density leads to about 13% increase in violent crime rate. Higher population density areas, especially those with the presence of wide-ranging commercial as well as non-commercial activities provide greater scope for crimes. As is evident from the results, a Democrat state has 4% lower crime rate compared to a Republican state.

In order to perform a robustness check to support the choice of the model, we include a year-fixed effect in the original linear regression model to absorb unaccounted year-specific effects. Results in column 2 of Table 6 and Table 7 suggest that the statistically significant variables are same for both the original model and the model including year-fixed effects. All four categories of expenditures exhibit results similar to the main model when the dependent variable is violent crime rate. Notably, several other control variables become statistically significant on including year-fixed effects in the model. Macroeconomic variable, inflation rate lead to an increase in both types of crimes as reflected in the existing literature (Rosenfeld, 2014). An increase in prices, strengthen the incentive to supply cheap, stolen goods in underground markets. Besides, increase in the price of goods (including essential items), may result in an increment of income-generating property crimes. As trade in stolen and/or illegal goods multiply beyond the control of formal authorities, involved parties may resort to violent means to attain their objectives, leading to an increase in violent crime rates. Increase in unemployment rate can only account for increase in property crime rates. This is consistent with Raphael and Winter-Ebmer (2001). Increase in unemployment rate implies the lack of opportunities to acquire the means to livelihood by legal economic activities which lowers the opportunity cost of property crime.

Variations in socioeconomic variable Gini Index can only explain violent crime rate. According to models of inequality and crime developed by Becker (1968) and Ehrlich (1973) low-income individuals with low expected return from legal market activity have a higher incentive to commit crime when in the vicinity of high-income individuals. The fraction of population who do not receive the fruits of economic growth are likely to engage in violent criminal activity. The demographic variable age dependency ratio marginally increases violent and property crime rates. An increase in age-dependency ratio implies an increase in the number of non-working individuals dependent on working individuals, either directly or via state welfare programs, for livelihood. This increases the economic burden on the existing workforce, which unless accompanied by economic opportunities may lead to increase in the occurrence of crimes. Further, a rise in economic burden when accompanied by psychological and physiological stress may also contribute to violent crimes.

Next, we estimate the system of Eqs. (1)–(5) using Generalized Method of Moments and the results are displayed in columns (3)–(7) of Table 6 and Table 7 for property crime rate and violent crime rate, respectively. Expenditures of welfare and correction are found to be statistically significant for both types of crimes while educational expenditure is significant for violent crime only. Hence, the crime-reducing effect of welfare expenditures is robust across all empirical models for property and violent crime and educational expenditure can explain violent crime rate in all three models. Similar to the previous model, the direction of change for correctional expenditures on crime rate is, unfortunately, positive and greater than the negative impact of welfare expenditures. For the remaining control variables, the results vary in some cases compared to the linear regression models of columns (1) and (2). It is possible to conclude that poverty rate and educational attainment influence both types of crime rates since these variables show consistent result across all models. Further, political orientation of the state is a significant explanatory variable for property crime and population density is that for violent crime across all three models.

Based on the GMM estimation results, we find no evidence of crowding out of one category of expenditure by another with one exception – expenditures on correction negatively affects expenditures on welfare. Thereby, it corroborates Marlow and Shiers (1999, 2001)'s finding that there is weak evidence of crowding out of spending on education by spending on crime-related programs. Additionally, it is interesting to note that while Republican states allocate higher percentage of GDP to education spending, a Democrat state allocates higher percentage of GDP to welfare spending. Welfare spending has a marginally greater impact on education. It is evident from Table 6 that property crime is lower in Democrat states. Further, spending in all categories are increasing functions of lagged GDP per capita with the exception of education spending. Additionally, variations in lagged crime rate do not explain variations in law enforcement spending, however, such variations contribute positively to correctional spending.

5. Conclusion and policy implications

This paper investigates the effectiveness of education and welfare spending vis-à-vis correctional and law enforcement spending as crime deterrents in the United States. Using panel data across 50 states during 1994–2014, we find that increasing spending on public welfare exerts a negative influence on both property and violent crime rates and is robust across all empirical models. However, the crime curbing effect of education is robust only for violent crime rate. Spending on correctional spending leads to increases in both types of crime rates. Among other macro and socioeconomic factors, an increase in poverty rate contributes to increase in violent and property crime rates while an increase in educational attainment leads to lowering of both types of crimes committed.

Based on the estimation results, it can be recommended that spending on welfare assistance be increased. Particularly, welfare programs that lead to provision of work relief that tend to lower unemployment rate would exhibit crime-deterring effect. Since we find strong evidence of a positive effect of poverty rate on crime, and some evidence that higher unemployment contributes to higher levels of crime, it further corroborates the conclusion that welfare programs targeted for the economically disadvantaged would be palliative to the problem of crime in society. Additionally, an increase in spending on education can potentially lead to lower crime rates as well as increased social and human capital. Even though the impact of education spending is not robust across both empirical models, the effect of educational attainment on crime is. Therefore, increasing education spending by the government that contributes to increase in the number of people with at least a bachelor's degree, would lower crime rates of both types. While there is no conclusive evidence on the effect of law-enforcement spending on crime rates, a reduction in correctional spending exhibits a positive effect on crime rates of both types. Given that the only evidence of crowding out was found in the case of welfare spending where increases in correctional spending led to reduction of welfare spending, along with the conclusion that welfare spending exhibits crime-reducing effect, a substitution of correctional spending by welfare spending is recommended.

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